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ODD END SEMESTER EXAMINATION DECEMBER 2024

Name of the Course: B. Tech(ECE)

Semester: VIIth Semester

Name of the Paper: Internet of Things and Its Applications Paper Code: TEC 759

Time: 3Hours

Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub-questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20marks)	
(a)	What are the characteristics of IoT that distinguish it from other network systems?	CO1
(b)	Explain the physical design of IoT, focusing on the role of 'Things' in IoT. What types of devices can be considered 'Things' in an IoT network?	
(c)	Describe the logical design of IoT. What are the functional blocks, and how do IoT communication models enable efficient data exchange?	
Q2	(20 marks)	
(a)	How does the IoT value chain reflect the complexity of interconnected systems, and what are its key components?	CO2
(b)	Describe the M2M value chain and explain how it contributes to the overall functioning of IoT systems.	
(c)	What are the main differences between Machine-to-Machine (M2M) communication and the Internet of Things (IoT)? How do their value chains differ?	
Q3	(20 marks)	
(a)	Describe the Functional View of the IoT Reference Architecture. What are the key functions, and how do they interact within an IoT system?	CO3
(b)	What is the Information View in the IoT Reference Architecture, and why is it crucial for data management in IoT systems?	
(c)	Outline the Deployment and Operational View in the IoT Reference Architecture. What considerations are necessary for the practical deployment of IoT systems?	
Q4	(20 marks)	
(a)	Discuss how IoT is revolutionizing the energy sector. What are the key applications in energy management, and how do they contribute to energy efficiency?	CO4
(b)	What are the major applications of IoT in agriculture, and how do they improve productivity and resource management in farming?	
(c)	How is IoT transforming healthcare and lifestyle management, and what are the key benefits and concerns related to IoT-enabled health devices?	
Q5	(20 marks)	
(a)	How are data platforms for smart cities addressing IoT security and privacy concerns, and what are the major challenges faced in this context?	CO5
(b)	Explain the Smartie Approach for secure IoT platforms. How does it enhance privacy, trust, and security in IoT data management?	CO5
(c)	What are the first steps towards developing a secure platform for IoT in smart cities, and how does data aggregation play a role in enhancing security and efficiency?	CO6

